space so that we don't miss out on a good region, but we also should focus on regions with high objective values to ensure that we are making progress in terms of optimization.

> **NOTE** **By "regions with high objective values", we mean the regions that comprise of inputs x that yield high values of outputs f(x), which are the target of our optimization (specifically maximization) task.**

Concretely, let's come back to our code example and assume that when we first start out, our training data set contains two observations from the Forrester objective function, at $x = 1$ and at $x = 2$:

```
train_x = torch.tensor([
    [1.],
    [2.]
])
train_y = forrester_1d(train_x)

model, likelihood = fit_gp_model(train_x, train_y)

print(torch.hstack([train_x, train_y.unsqueeze(1)]))
```

which gives the output:

```
tensor([[1.0000, 1.6054],
        [2.0000, 1.5029]])
```

This indicates that the point at 1 evaluates at roughly 1.6, while the point at 2 evaluates at 1.5. Visualizing the predictions made by the trained GP, we obtain the familiar-looking plot in Figure 4.6. This figure shows the exploration–exploitation tradeoff we face: should we evaluate the objective function where uncertainty is high, or should we stay within the region where the mean prediction is high?

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